Uncertainty and Sentiment in ARK Company Filings

Yuanpeng Chen | NetID yc8027 | Fall 2026

Repository: github.com/victor4628/FRE7871-NLP

I measure negative language and uncertainty in 1,536 10-K/10-Q filings from 91 ARK-universe issuers during 2021-2025. I compare proportional counts with Loughran-McDonald TF-IDF, test form-specific trends, then relate uncertainty to subsequent volatility and negative language to the four-day filing return.

Table 1 Sample construction

Security and issuer screen	Removed	Remain
Unique cleaned securities in frozen snapshot	0	124
SEC ticker-to-CIK mapping found	7	117
At least one 2021-2025 10-K/10-Q	24	93
Unique companies (GOOG and GOOGL count as one)	1	92

Unique-filing screen	Removed	Remain
Starting unique filings	0	1682
Parsed non-amended 10-K/10-Q	0	1682
At least 2,000 words (10-K) or 1,000 words (10-Q)	0	1682
Earliest filing per company-calendar quarter	14	1668
Usable day 0 and day -1 price at least $3	108	1560
At least 60 return days before and after day 0	23	1537
Complete share count, controls and outcomes	1	1536

The 1,702 security-filing records include 20 duplicate Alphabet records; retaining GOOGL leaves 1,682 unique filings. The form-specific word thresholds remove none. The company-quarter rule removes 14, the $3 price rule removes 108, the 60-day history rule removes 23, and one JOBY filing lacks a usable filing-date share count. No filing failed parsing.

Day 0 is the first exchange session on or after the later of the filing date and the SEC acceptance date, shifted one day when acceptance is at or after 16:00 Eastern. This moves 860 retained filings. Shares come from the same filing; common classes are summed only when no consolidated total exists. Visible inline-XBRL text is retained, hidden material is removed, and tables above 15% digits are stripped.

Word lists and method

The LM lists contain 2,355 Negative and 297 Uncertainty words, overlapping on 40 words. Proportions divide list-word occurrences by all tokens. For term i in document j, equation (1) is:

$w_{ij} = [(1 + \ln tf_{ij}) / (1 + \ln a_j)] \ln(N / df_i)$ for tf > 0; otherwise zero, where a_j is total tokens divided by distinct tokens. Category TF-IDF is the sum of its term weights. Natural logs are used.

The repository self-check gives TF-IDF values 0.8480, 0.2885 and 0.5026 for documents d1-d3. IDF is recomputed on each regression corpus. The quarterly chart first averages within issuer, form and filing quarter, then weights issuers equally. Table 4 adds seasonal effects; its preferred within-company model also adds issuer effects and clusters standard errors by issuer. Aggregate Newey-West errors use four lags because quarterly residuals may be serially correlated.

Tables 5-6 include log filing-date market value, log average dollar volume and SPY excess return over [-60,-6], a 10-K indicator, company effects and calendar-quarter effects. Table 5 compares models without and with log pre-filing volatility. Volatility is annualized from [-60,-6] before and [+4,+63] after. Table 6 uses stock minus SPY buy-and-hold return over [0,+3], measured from the day -1 close. Outcome-model standard errors cluster by company.

Table 2 Summary statistics by form

Form / measure	N	Mean	SD	P25	Median	P75
10-K / Negative %	381	2.318	0.413	2.045	2.344	2.582
10-K / Negative TF-IDF	381	163.515	56.188	119.102	149.910	199.653
10-K / Uncertainty %	381	1.935	0.241	1.790	1.953	2.114
10-K / Uncertainty TF-IDF	381	26.685	8.484	20.720	25.281	31.680
10-Q / Negative %	1,155	2.108	0.940	1.266	1.890	2.993
10-Q / Negative TF-IDF	1,155	79.247	65.775	24.322	55.471	127.759
10-Q / Uncertainty %	1,155	1.808	0.582	1.317	1.644	2.405
10-Q / Uncertainty TF-IDF	1,155	13.471	7.275	7.874	11.925	18.467

Negative-uncertainty correlations are 10-K: 0.731 proportional and 0.851 TF-IDF; 10-Q: 0.846 and 0.920.

Table 3 Words driving each measure

Each share divides a word count by all occurrences on its own list. The two lists appear side by side.

Rank	Negative word	List share	Uncertainty word	List share
1	LOSS	5.16%	MAY	34.79%
2	ADVERSELY	4.08%	COULD	18.35%
3	LOSSES	3.18%	RISK	5.05%
4	CLAIMS	3.01%	RISKS	4.53%
5	ADVERSE	2.72%	BELIEVE	2.95%
6	AGAINST	2.42%	APPROXIMATELY	2.65%
7	LITIGATION	1.94%	ASSUMPTIONS	1.91%
8	UNABLE	1.92%	INTANGIBLE	1.70%
9	HARM	1.86%	POSSIBLE	1.32%
10	FAILURE	1.79%	UNCERTAINTIES	1.16%
11	FAIL	1.32%	FLUCTUATIONS	1.14%
12	IMPAIRMENT	1.24%	MIGHT	1.09%
13	NEGATIVELY	1.14%	UNCERTAIN	1.03%
14	DIFFICULT	1.13%	ANTICIPATED	0.98%
15	PENALTIES	1.13%	VOLATILITY	0.94%
16	CRITICAL	1.02%	DEPEND	0.94%
17	NEGATIVE	1.02%	PREDICT	0.94%
18	DAMAGES	0.92%	UNCERTAINTY	0.89%
19	DECLINE	0.91%	DIFFER	0.86%
20	DELAYS	0.91%	ANTICIPATE	0.80%
21	RESTATED	0.89%	EXPOSURE	0.79%
22	LIMITATIONS	0.86%	CONTINGENT	0.78%
23	DELAY	0.81%	VARIABLE	0.71%
24	VOLATILITY	0.77%	DEPENDS	0.68%
25	CLOSING	0.76%	PENDING	0.65%
26	CHALLENGES	0.76%	DEPENDENT	0.58%
27	FINES	0.71%	CONTINGENCIES	0.56%
28	HARMED	0.69%	PROBABLE	0.52%
29	DISRUPTIONS	0.69%	ASSUMED	0.52%
30	INVESTIGATIONS	0.67%	VARY	0.49%

Q1. What is each measure actually made of? The ten leading words supply 28.1% of Negative counts and 74.4% of Uncertainty counts; the top 30 supply 46.5% and 90.3%. MAY and COULD alone supply 53.1%. Together with RISK/RISKS, this looks mainly like recurring risk-factor language rather than filing-specific managerial hedging. I did not estimate sentence-level duplication, so “copied forward unchanged” remains an interpretation.

Q2. Are sentiment and uncertainty measuring different things? Pooled correlations are 0.841 for proportions and 0.919 for TF-IDF. PYPL's 2024 10-K is a contrast: its Negative proportion is at the 89th percentile but Uncertainty at the 25th, driven by losses, fraud, regulatory penalties and litigation. The high correlations mean much of the report is one shared disclosure-intensity result; differing trends and outcome coefficients show the measures are related, not identical.

Figure 1 Language scores and VIX

Annual and quarterly reports remain separate; issuer weighting prevents frequent filers from dominating. VIX is a market comparison, not the firm-volatility outcome.

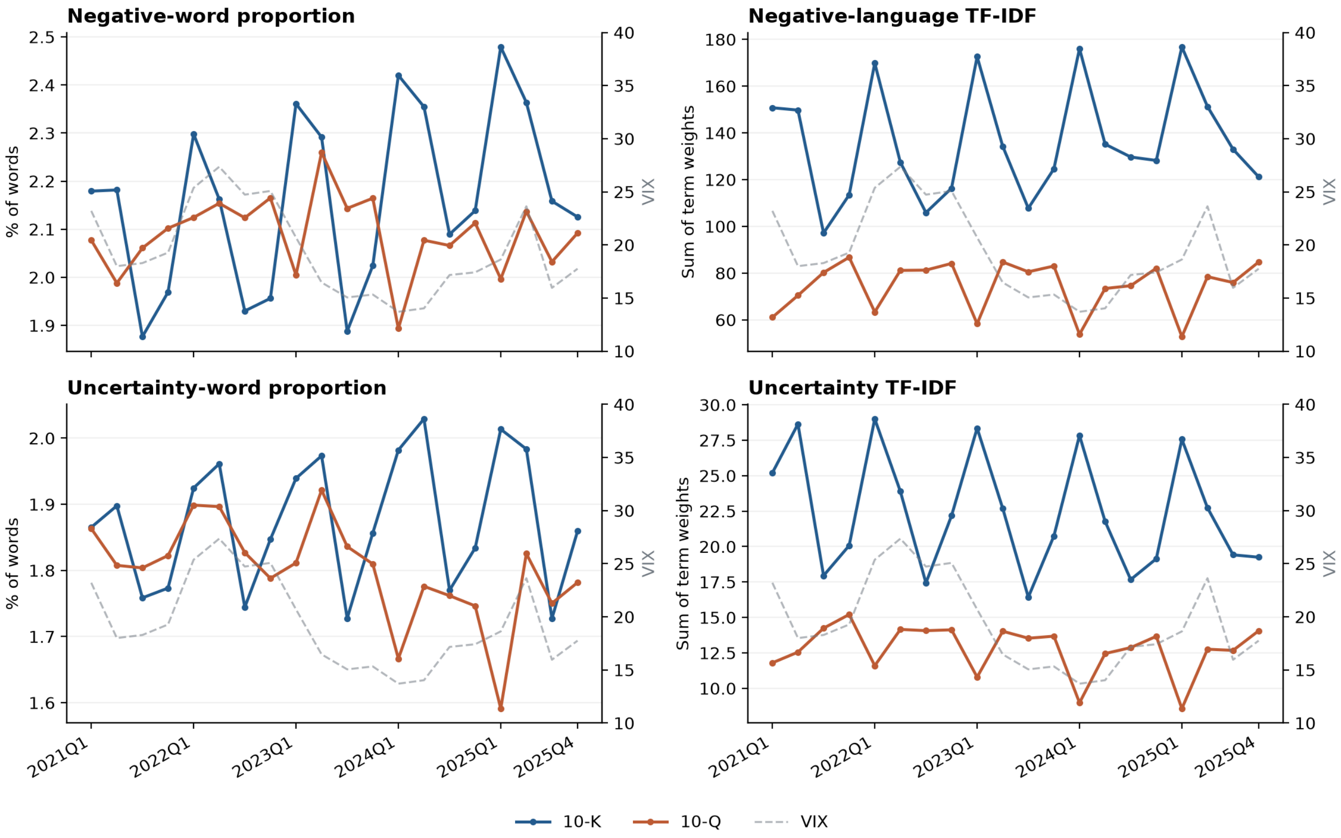

Table 4 Trends, aggregate and within company

Form	Measure	Overall change/year	OLS t	NW t	Within-company change/year	Within t	Within p
10-K	Negative %	0.062	9.15	12.83	0.063	7.21	<0.001
10-K	Negative TF-IDF	2.892	3.27	5.85	2.630	3.78	<0.001
10-K	Uncertainty %	0.018	3.52	3.20	0.025	5.16	<0.001
10-K	Uncertainty TF-IDF	-0.194	-1.01	-1.51	-0.029	-0.26	0.797
10-Q	Negative %	-0.009	-0.77	-0.81	-0.007	-0.45	0.651
10-Q	Negative TF-IDF	-1.345	-1.85	-1.99	-1.519	-1.26	0.211
10-Q	Uncertainty %	-0.029	-2.88	-3.05	-0.020	-1.97	0.052
10-Q	Uncertainty TF-IDF	-0.516	-3.97	-4.64	-0.482	-3.46	<0.001

Proportional changes are percentage points per year; TF-IDF changes are term-weight units per year. Aggregate regressions include seasonal effects. Within-company regressions include issuer and seasonal effects and issuer-clustered errors.

Q3. Did either measure trend over 2021-2025? I trust the within-company model because it removes stable company differences and accounts for repeated filings. In 10-Ks, Negative rises 0.063 percentage points/year (t=7.21) and 2.630 TF-IDF units/year (t=3.78). Negative does not trend in 10-Qs (-0.007 points/year, t=-0.45; TF-IDF t=-1.26). 10-K Uncertainty rises proportionally (0.025 points/year, t=5.16) but not with TF-IDF (t=-0.26), while 10-Q Uncertainty falls most clearly under TF-IDF (-0.482 units/year, t=-3.46). The evidence supports within-company change for 10-K negative language and 10-Q uncertainty, rather than a single pooled trend.

Table 5 Uncertainty and subsequent volatility

The dependent variable is log annualized volatility on [+4,+63]. Both columns contain all controls listed on page 2; the second additionally includes log volatility on [-60,-6]. Scores are standardized within the same 1,536-filing sample. Parentheses contain company-clustered standard errors.

Proportion

Variable / statistic	Without prior volatility	With prior volatility
Uncertainty score (+1 standard deviation)	0.0119 (0.0156)	0.0003 (0.0130)
Prior volatility (log)	-	0.3698 (0.0964)
Score p-value	0.446	0.984
Number of filings	1,536	1,536

TF-IDF

Variable / statistic	Without prior volatility	With prior volatility
Uncertainty score (+1 standard deviation)	0.0336 (0.0257)	0.0136 (0.0176)
Prior volatility (log)	-	0.3682 (0.0956)
Score p-value	0.193	0.443
Number of filings	1,536	1,536

Q4. Does uncertainty language predict volatility? Adding prior volatility changes the proportional coefficient from 0.0119 to 0.0003, a 97.8% attenuation. TF-IDF falls from 0.0336 to 0.0136, a 59.7% attenuation. Neither controlled coefficient is precise. Existing volatility explains most of the proportional association and much of the TF-IDF association; uncertainty does not robustly predict a change in volatility.

Q5. Do 10-Qs behave like 10-Ks?

Form	Weight	Without prior vol: coefficient / p	With prior vol: coefficient / p
10-K	Proportion	-0.030 / 0.819	-0.032 / 0.796
10-K	TF-IDF	0.226 / 0.062	0.194 / 0.041
10-Q	Proportion	0.046 / 0.151	0.008 / 0.714
10-Q	TF-IDF	0.072 / 0.062	0.026 / 0.280

10-Q proportional scores vary more despite shorter, more templated reports: Negative SD is 0.940% versus 0.413% for 10-K, and Uncertainty SD is 0.582% versus 0.241%. A shorter denominator makes proportions more sensitive to section mix and episodic wording. Because a 10-Q usually follows an earnings release by only days, daily filing returns cannot separately identify the report from earnings news. Longer 10-Ks should carry richer textual signal, but Table 5 is mixed: only 10-K TF-IDF is significant with prior volatility, while its proportional counterpart and both 10-Q estimates are not. That isolated result is insufficient to establish a form difference.

Table 6 Negative language and four-day excess returns

The dependent variable is stock minus SPY buy-and-hold return over [0,+3], in percentage points. Controls and fixed effects are those listed on page 2; standard errors cluster by company.

Weight	Coefficient	SE	t	p	80% MDE	N
Proportion	-0.689	0.597	-1.15	0.252	1.69	1,536
TF-IDF	-0.862	0.728	-1.18	0.239	2.06	1,536

Replacing SPY with ARKK gives -0.509 (p=0.368) and -0.521 (p=0.445).

Q6. Which results do I believe? I believe the 10-K Negative trend because both weights agree and the preferred within-company t-statistics are 7.21 and 3.78. I give less weight to Uncertainty trends because 10-K proportion and TF-IDF disagree; the 10-Q decline is more credible under TF-IDF but only borderline under proportions. I believe the volatility comparison: adding prior volatility materially shrinks both coefficients, and the remaining pooled estimates are imprecise. I do not treat the isolated 10-K TF-IDF result as robust. The return coefficients are negative under both weights, but p-values are 0.252 and 0.239 and the approximate 80%-power minimum detectable effects are 1.69 and 2.06 percentage points per SD. This test is underpowered, so it supports neither return predictability nor a zero effect.

Limitations and next step

The universe is selected from 2026 holdings snapshots. Only 71 of 91 retained issuers appear in every filing year; firms that failed, disappeared or left ARK before the snapshot cannot enter. Twenty aggregate quarters limit time-series inference. Full-corpus IDF is retrospective, daily timing cannot separate nearby news, and SPY is an imperfect benchmark; the ARKK check does not change the return conclusion.

The single best improvement is to reconstruct historical ARK membership and evaluate scores out of sample. It would require dated holdings archives and rebuilding every filing-level sample by the company's actual inclusion date.

Sources: Loughran and McDonald (2011), Journal of Finance 66(1), 35-65; Fall 2026 assignment brief; instructor repository; LM dictionary; Yahoo Finance; exchange_calendars; statsmodels. AI assistance is disclosed in AI_USE.md.